# AI-Assisted Research into Alternative Life: An Evidence-Grounded Computational Framework for Oxygen-Independent Metabolism and Hypothetical Biochemistries

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## Abstract

**Background.** Research on alternative life asks which features of familiar biology are universal and which are accidents of Earth's history. Artificial intelligence (AI) can speed up literature synthesis and hypothesis generation in this field, but language models can fabricate citations, omit caveats and propose chemically impossible reactions. **Problem.** Public discussion often conflates two very different claims: that an organism does not require molecular oxygen ($O_2$), which is well established for anaerobes, and that an organism contains no oxygen atoms at all, which is speculative and undemonstrated. **Approach.** This thesis proposes a reproducible, human-supervised workflow that combines (i) source-linked evidence extraction, (ii) a four-level evidence taxonomy, (iii) constraint-based metabolic modelling with COBRApy, (iv) automated atom- and charge-balance checks, and (v) an uncertainty-aware feasibility rubric. Known oxygen-independent metabolism serves as the benchmark domain. **Evaluation.** A pre-specified comparison of an unstructured AI summary against the evidence-grounded workflow is defined, with metrics for citation support, unsupported-claim rate, error detection, reviewer agreement, reproducibility and usefulness. **Contribution.** The contribution is a transparent method and evaluation design, not the discovery of a new organism. The framework keeps claims proportional to evidence by separating what is demonstrated, what is inferred and what is speculative.

**Keywords:** artificial life; astrobiology; anaerobic metabolism; artificial intelligence; computational biology; flux balance analysis; xenobiology; hypothesis generation; research reproducibility

## 1. Introduction

### 1.1 Background

Earth hosts organisms in oxygen-rich surface environments and in anoxic habitats such as sediments, the deep subsurface and some hydrothermal systems. Some organisms require $O_2$, whereas others conserve energy through fermentation or respire using other electron acceptors. This diversity raises a basic question: which properties of life are universal, and which reflect the particular evolutionary history of one planet (Pace, 2001)? Artificial-life research explores life-like behaviour in simulations, engineered biological systems and hypothetical chemistries (Langton, 1997; Stepney, 2025). Xenobiology goes further by building biological systems with non-natural components, such as synthetic genetic polymers that can store information and evolve (Pinheiro et al., 2012). Astrobiology asks how varied solvents, energy sources and elemental compositions could support living systems (Bains, 2004).

AI systems are increasingly used to connect evidence across biological research fields (Qi et al., 2026; Rao et al., 2026). Their outputs, however, are only as trustworthy as the evidence retrieval, validation and oversight surrounding them.

## 1.2 Two meanings of "without oxygen"

The phrase "life without oxygen" can mean two very different things, and a third level sits between them. Table 1 separates the claims.

| Level | Claim | Status |
|---|---|---|
| 1 | A process or organism does not require molecular oxygen ($O_2$) | Established. Anaerobic fermentation and respiration are documented. |
| 2 | An organism does not use $O_2$ as terminal electron acceptor but still uses oxygen-containing molecules (for example nitrate, sulfate, $CO_2$, metal oxides, water) | Established examples exist. Some anaerobes use alternative electron acceptors but still contain oxygen-bearing molecules. |
| 3 | An organism's molecules contain no oxygen atoms at all | Speculative. No such organism is known; many familiar biomolecules contain oxygen. |

*Table 1. Three levels of the claim "life without oxygen".*

The existence of anaerobic organisms supports Level 1 and provides examples of Level 2. Neither establishes Level 3. Many anaerobic electron acceptors and metabolites contain oxygen atoms, illustrating that the absence of $O_2$ gas does not imply oxygen-free molecular composition.

## 1.3 Problem statement

AI tools can generate plausible hypotheses about unfamiliar life, but their scientific value is uncertain unless the hypotheses are connected to verifiable sources, formal chemical constraints and explicit uncertainty. The problem addressed here is how to design a workflow that explores alternative-life hypotheses without conflating established anaerobic biology with unsupported claims about oxygen-atom-free organisms.

## 1.4 Research questions

**Main question (RQ).** How can an AI-assisted computational workflow improve the traceability, consistency and scientific usefulness of hypotheses about alternative life, using known oxygen-independent metabolism as a benchmark?

**Sub-questions.**

- **RQ1.** Can AI-assisted extraction produce an evidence map in which every claim is linked to a verifiable source passage?
- **RQ2.** Can metabolic-network analysis distinguish pathways that do not require $O_2$ from claims that oxygen-containing chemistry can be eliminated?
- **RQ3.** Which checks (atom balance, charge balance, thermodynamic plausibility, citation verification) catch errors that language-model review alone misses?
- **RQ4.** How should a workflow separate established observations, model-based inferences, speculation and unsupported claims?
- **RQ5.** What evidence would be required before a proposed alternative biochemistry could be considered chemically and biologically plausible?

## 1.5 Research gap and contribution

Research on AI for biological discovery, computational metabolic modelling and alternative biochemistry offers complementary methods, but their integration requires explicit safeguards for evidence provenance and chemical consistency. This manuscript proposes a workflow joining AI-supported exploration with source-level evidence and formal checks. Its proposed contributions are:

1. A conceptual separation of three levels of "oxygen independence" (Table 1).
2. A four-class evidence taxonomy and an auditable evidence-record schema.
3. A validation layer combining constraint-based modelling with atom- and charge-balance checks.

4. A feasibility rubric for speculative alternative-life claims.
5. A pre-specified, blinded evaluation protocol that can be executed and reported honestly, including failures.

## 2. Background and Related Work

### 2.1 Anaerobic metabolism

Anaerobic organisms obtain energy by fermentation or by respiration with electron acceptors other than $O_2$, depending on organism and environment. Their existence shows that $O_2$ gas is not universally required for energy metabolism. It does not show that oxygen atoms can be removed from cellular chemistry. Metabolic pathways can be represented as networks of metabolites and reactions, with stoichiometric matrices encoding the quantities of reactants and products.

### 2.2 Artificial life and xenobiology

Artificial-life research studies life-like systems and the principles that might allow life to arise or persist in different substrates (Langton, 1997; Stepney, 2025). Xenobiology studies systems whose chemistry differs from terrestrial biology, including alternative nucleic-acid backbones (Pinheiro et al., 2012). Theoretical work on synthetic life draws lessons from origin-of-life research (Peng, 2023). This literature shows both flexibility and constraint: molecular stability, replication, catalysis, energy transfer and whole-system compatibility remain demanding requirements. A simulation of life-like behaviour is not, by itself, evidence that the simulated system is alive.

### 2.3 Alternative biochemistries

Bains (2004) argued that many chemistries could in principle support living systems, while also noting that each proposal faces constraints from stability, solvent properties and energetics. Proposals for non-aqueous solvents or non-oxygen-based building blocks are therefore best treated as hypotheses requiring chemical and thermodynamic scrutiny rather than as established possibilities.

### 2.4 AI for biological research and its failure modes

AI systems can assist with literature search, data analysis, hypothesis generation and iterative workflows (Qi et al., 2026), and generalist biological models are advancing rapidly (Rao et al., 2026). Known failure modes include hallucinated or mis-attributed citations (Ji et al., 2023), weak handling of uncertainty, incomplete knowledge and difficulty linking statistical patterns to causal mechanism. This thesis therefore treats AI output as a candidate to be verified, never as an authority.

### 2.5 Constraint-based metabolic modelling

Constraint-based reconstruction and analysis (COBRA) models metabolism using mass-balance constraints and reaction bounds. Flux balance analysis (FBA) estimates flux distributions that satisfy those constraints while optimising an objective such as growth (Orth et al., 2010). COBRApy is an open-source Python implementation (Ebrahim et al., 2013). Curated genome-scale models, such as the *Escherichia coli* model iJO1366 (Orth et al., 2011), are available through repositories such as BiGG Models. Results depend on network completeness, reaction bounds and modelling assumptions. Setting $O_2$ uptake to zero tests dependence on molecular oxygen under those assumptions only. It cannot establish oxygen-atom-free life.

## 3. Conceptual Framework

### 3.1 Definitions

- **Alternative life:** life-like systems that differ from known terrestrial organisms in substrate, chemistry, environment or implementation.
- **Anaerobic:** not requiring molecular oxygen for a specified process or growth condition.

- **Oxygen-atom-free:** a system whose constituent molecules contain no oxygen atoms. No living example is known.
- **AI-assisted:** AI supports retrieval, classification, synthesis or hypothesis generation, while evidence checks and human judgment remain essential.

### 3.2 Evidence taxonomy

Every claim handled by the proposed workflow is assigned exactly one evidence class (Table 2).

| Class | Label | Meaning | Example |
|---|---|---|---|
| A | Established observation | Directly supported by empirical evidence in a verified source | Some microbes grow without $O_2$ |
| B | Inference | Follows from a model or theory under stated assumptions | A model grows with $O_2$ uptake bounded at zero |
| C | Speculation | A hypothesis with no direct supporting evidence | A cell built entirely without oxygen atoms |
| D | Unsupported or contradicted | Not supported by, or conflicting with, the evidence | A fabricated citation; an atom-imbalanced reaction |

*Table 2. Evidence classes.*

The system must be allowed to return "insufficient evidence". A claim that fails validation is corrected or marked unresolved, never silently retained.

### 3.3 Feasibility rubric for speculative claims

Claims in Class C are scored against five criteria, each rated *met*, *partly met*, *not met* or *unknown*, with a written rationale.

| Criterion | Question |
|---|---|
| Chemical validity | Are all proposed reactions atom- and charge-balanced? |
| Thermodynamic plausibility | Is the overall free-energy change favourable under stated conditions, using reliable data? |
| Stability and environment | Are the molecules stable in the proposed solvent, temperature and pressure range? |
| Functional completeness | Are information storage, catalysis, compartmentalisation and energy conservation all addressed? |
| Empirical accessibility | Is there a feasible observation or experiment that could support or refute the claim? |

*Table 3. Feasibility rubric.* A claim can score well on chemical validity and still be biologically impossible or kinetically inaccessible. The rubric records such gaps rather than hiding them.

## 4. Methodology

### 4.1 Research design

The proposed study follows a design-science approach combining a scoping literature review protocol with a computational proof of concept. Four phases are planned: (1) evidence collection, (2) AI-assisted claim extraction with human verification, (3) computational validation using documented metabolic models and balance checks, and (4) evaluation against predefined criteria. The review and computational analysis remain to be executed; no wet-laboratory work is required.

### 4.2 Literature search and screening

To execute the proposed scoping review, search PubMed, Google Scholar and relevant publisher platforms using combinations of "anaerobic metabolism", "artificial life", "xenobiology", "alternative biochemistry", "AI for scientific discovery" and "constraint-based metabolic modelling". Record the search date, exact query, database, screening decision and reasons for exclusion. Prefer peer-reviewed reviews, primary research and official software documentation, and verify each source directly rather than relying on search snippets. Maintain a screening table containing title, authors, year, DOI or stable URL, publication type, key claim, supporting passage, evidence class and inclusion decision. Popular summaries that cannot be traced to primary sources should be excluded from the core evidence set. Because the search has not yet been run and documented, this manuscript does not claim to be a completed systematic or scoping review.

### 4.3 Evidence extraction

For each included source, claims are extracted into structured records (Appendix A). AI may suggest summaries, but a human reviewer must check each claim against its source passage. Exact source passages and identifiers are stored for auditability, and model-generated text is stored separately from verified source text. Generated citations are never added to the bibliography without independent verification.

### 4.4 Computational modelling

The planned computational analysis will use Python and COBRApy to load a published, well-documented model suitable for analysis (for example, iJO1366 from BiGG Models). The procedure is:

1. Reproduce a baseline FBA analysis under the model's documented conditions.
2. Constrain $O_2$ uptake to zero and re-optimise the model objective. Report the objective value and relevant fluxes, noting that a feasible solution is not necessarily evidence of biologically realistic growth; the interpretation depends on model structure and constraints.
3. Record model identifier and version, source database, software versions, objective function, exchange constraints, reaction bounds, solver and every modification.
4. Check mass balance and reaction annotations.

This tests an existing model and is not a design for a novel organism. A zero-$O_2$-uptake constraint concerns molecular oxygen only and says nothing about oxygen-atom-free chemistry.

### 4.5 Atom- and charge-balance validation

For benchmark reactions, a script should parse each chemical formula and compare elemental totals on both sides; charge should also be checked when reaction charges are explicitly represented. The following are illustrative worked examples, not results from the proposed study.

| Reaction | C | H | O | Verdict |
|---|---|---|---|---|
| $C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$ | 6 → 4+2 = 6 | 12 → 12 | 6 → 2+4 = 6 | Balanced |
| $C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + CO_2$ | 6 → 4+1 = 5 | 12 → 12 | 6 → 2+2 = 4 | Imbalanced (C and O) |

*Table 4. Illustration of elemental balance checking (ethanol fermentation of glucose).*

Balance is necessary but not sufficient. A balanced equation may still be thermodynamically unfavourable, kinetically inaccessible or incompatible with a functioning cell. Where reliable thermodynamic data exist, free-energy feasibility is assessed separately.

### 4.6 Hypothesis generation and validation

AI is used to propose research questions and competing explanations. Each hypothesis record contains the claim, rationale, assumptions, supporting sources, counter-evidence, testable implication, required data and uncertainty.

Hypotheses then pass through the validation layer (citation check, balance check, rubric scoring) and human review before being reported.

### 4.7 Comparison design

Two workflows are compared on identical sources and task prompts:

- **Baseline (W1):** an AI-generated summary without mandatory evidence structure.
- **Evidence-grounded (W2):** the proposed workflow, requiring source passages, evidence classes and formal checks.

Two reviewers independently score a sample, blind to workflow condition where practical, and resolve disagreements through a documented procedure.

### 4.8 Hypotheses to be tested

- **H1.** Requiring every AI-generated claim to link to a source passage lowers the unsupported-claim rate relative to W1.
- **H2.** Adding stoichiometric and atom-balance checks identifies chemically inconsistent reaction hypotheses missed by language-model review alone.
- **H3.** Separating claims into evidence classes improves inter-reviewer agreement about evidence strength.

These are hypotheses, not findings. They require a defined benchmark, a documented sampling procedure and the scoring protocol in Section 6 before they can be tested.

### 4.9 Evaluation metrics

| Metric | Definition |
|---|---|
| Citation support rate | Proportion of checked claims whose cited passage genuinely supports the claim |
| Unsupported-claim rate | Proportion of claims judged Class D |
| Balance-check detection | Invalid reactions correctly flagged in a documented test set (sensitivity and false-positive rate) |
| Reviewer agreement | Percentage agreement and Cohen's kappa (Cohen, 1960) for evidence classes |
| Reproducibility | Proportion of runs reproducible from saved code, data, parameters and environment |
| Usefulness | Reviewer ratings of clarity, relevance and testability on a predefined rubric |

*Table 5. Evaluation metrics.*

### 4.10 Data analysis

Report counts and proportions with uncertainty intervals where appropriate. For small exploratory samples, prefer transparent descriptive analysis to unsupported claims of statistical significance. Show successful and failed cases. Share the search log, extraction schema, rubric, model configuration and code where licensing and institutional rules permit. These are planned reporting requirements, not completed results.

## 5. Proposed System Architecture

### 5.1 Components

- **Literature retriever:** searches scholarly sources and records bibliographic metadata.
- **Evidence store:** retains source passages, claim identifiers, evidence classes and provenance.
- **AI synthesis module:** summarises retrieved evidence and proposes competing hypotheses.
- **Metabolic analysis module:** loads an existing network model and runs defined constraint-based analyses.

- **Validation layer:** checks citation support, elemental and charge balance, schema completeness and label consistency.
- **Human review interface:** lets a researcher accept, edit, reject or mark claims uncertain.
- **Reporting module:** generates a traceable report of methods, assumptions and limitations.

### 5.2 Workflow

Define question → retrieve sources → screen sources → extract claims → assign evidence class and uncertainty → generate candidate hypotheses → run computational checks → human review → report.

### 5.3 Implementation stack

Python, Jupyter, pandas, Git, a reference manager and COBRApy. Claims and provenance can be stored in structured JSON files or a lightweight database. The workflow must remain usable if the AI component is replaced by manual review. Software versions are pinned and random seeds saved.

## 6. Evaluation Protocol

### 6.1 Benchmark tasks

1. Classify claims about anaerobic metabolism as supported, contradicted or unresolved, using selected papers.
2. Summarise a known metabolic pathway with key statements linked to evidence.
3. Distinguish "does not require molecular oxygen" from "contains no oxygen atoms" (Table 1).
4. Detect atom- or charge-imbalanced reaction equations in a transparent test set.
5. Produce ranked research questions with assumptions, evidence gaps and proposed tests.

### 6.2 Procedure and reporting

The benchmark must be created and frozen before system comparison. Reviewers should establish reference labels from source documents, and both workflows should receive the same tasks and source set. Save prompts, model versions, settings and outputs. Score outputs blind to workflow condition where practical; predefine how ambiguous cases are resolved and do not select examples after observing performance. After execution, report each metric, both workflow scores, sample size, scoring method, uncertainty and failure cases. No evaluation has yet been run for this manuscript. Success is defined as a reproducible workflow and transparent evaluation, not discovery of a new organism.

## 7. Expected Contributions and Discussion

### 7.1 Methodological contribution

The framework offers a practical process for combining AI-supported exploration with source-level evidence and formal checks. It helps interdisciplinary researchers avoid presenting speculation as established science.

### 7.2 Scientific contribution

The three-level separation in Table 1 clarifies what anaerobic organisms demonstrate and what they do not. An organism may avoid molecular oxygen in a given process while still containing oxygen atoms in many molecules; some anaerobic metabolisms also use oxygen-bearing electron acceptors.

### 7.3 Implications for astrobiology and AI

Alternative-life reasoning can broaden how astrobiology considers solvents, energy sources and substrates. Models must still respect thermodynamics, chemical availability, environmental stability and the difficulty of recognising

unfamiliar life. AI systems in this setting should be judged by evidence fidelity, reproducibility and testability, not by confident language.

## 8. Limitations and Threats to Validity

- Published knowledge may be incomplete or biased toward well-studied organisms.
- Language models may hallucinate references, misread context or miss contradictions. Human verification reduces but does not remove this risk.
- Metabolic models may omit regulation, spatial effects, kinetics, unknown reactions or environmental constraints.
- A zero-$O_2$-uptake simulation cannot answer whether all oxygen atoms can be removed from a living system.
- Life has no universally accepted operational definition for every virtual or hypothetical case.
- A small exploratory evaluation, with few reviewers, may not generalise to other domains or AI systems, and reviewer expectations may bias scoring.
- No experimental evidence for oxygen-atom-free life is presented, and no feasibility claim is made.

## 9. Ethics, Safety and Reproducibility

The proposed computational project uses public literature and appropriately licensed models and does not require creating or releasing novel organisms. Any future laboratory extension must receive appropriate institutional review and comply with applicable biosafety, biosecurity and ethics requirements.

Declaration of Generative AI Assistance: Generative AI tools assisted with literature retrieval structuring and initial stylistic drafting of this protocol. The author independently audited and verified all primary literature references, chemical formulas, stoichiometric calculations, and conceptual claims. The author accepts full scientific responsibility for the final contents of this work.

**Reproducibility checklist.** Preserve search queries and dates; retain bibliographic identifiers; version-control code; document software versions; record model files and parameters; save raw evaluation outputs; predefine metrics; report failures and uncertainty.

## 10. Conclusion and Future Work

AI-assisted research into alternative life is a promising interdisciplinary topic linking computer science, chemistry, biology and astrobiology. Its credible near-term contribution is a reliable workflow for exploring hypotheses, testing models and identifying knowledge gaps, not the invention of a new organism. Anaerobic organisms show that molecular oxygen is not necessary for every form of energy metabolism. They do not show that oxygen atoms are unnecessary for life as we know it, and oxygen-atom-free life remains a speculative chemical question.

This manuscript defines a human-supervised workflow combining literature retrieval, evidence extraction, metabolic modelling, formal checks and transparent evaluation. Its value depends on implementation and measured performance, which remain future work. Natural extensions include additional metabolic models, comparison of AI systems, stronger chemical-reasoning tools, thermodynamic feasibility modules and expert review of generated hypotheses. Claims should distinguish possibility, plausibility and demonstrated fact.

## Data, Code and Protocol Availability

Data, Code, and Protocol Availability: All computational tools, balance validation scripts, curated evidence records, benchmark datasets, and protocol documentation are open-source and publicly accessible in the project GitHub repository: https://github.com/Zainul9142/AI-Assisted-Research-into-Alternative-Life. The permanent archival record of this framework is deposited on Zenodo under DOI: 10.5281/zenodo.23574418 (https://doi.org/10.5281/zenodo.23574418).

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## Appendix A. Evidence-Record Schema

Each claim record contains: `claim_id`; `claim_text`; `source_title`; `authors`; `year`; `doi_or_url`; `access_date`; `supporting_passage` (verbatim, stored separately from paraphrase); `evidence_class` (A–D); `counterevidence`; `reviewer_status` (accepted, edited, rejected, uncertain); `confidence_rationale`; `validation_checks`; and `notes`.

## Appendix B. Twelve-Week Work Plan

| Weeks | Activities |
|---|---|
| 1–2 | Study basic metabolism, anaerobic respiration, artificial life and research methods |
| 3–4 | Search and screen literature; build the evidence database |
| 5–6 | Set up Python tools; reproduce a documented metabolic-model analysis |
| 7–8 | Implement claim provenance, evidence classes and consistency checks |
| 9–10 | Build the benchmark, compare workflows, collect reviewer scores |

| Weeks | Activities |
|---|---|
| 11–12 | Analyse outputs, document limitations, write the report, obtain supervisor review |